A_∞-Chiral Algebras and Three-Dimensional Holomorphic–Topological Field Theory

The Companion

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Abstract

The BV/BRST quantization of a three-dimensional holomorphic–topological field theory on $\mathbb{C}_z \times \mathbb{R}_t$ is the chiral bar construction of its boundary chiral algebra. Holomorphic collisions on $\mathbb{C}$ generate the bar differential; topological point-splittings on $\mathbb{R}$ generate the coproduct; the two-coloured Swiss-cheese chiral–topological operad $\mathsf{SC}^{\mathrm{ch,top}}$ is the universal home of the bulk–boundary structure. Under this identification the quantum master equation is the Maurer–Cartan equation, each A_∞ identity is Stokes' theorem on a Fulton–MacPherson compactification, and the conformal anomaly is bar curvature: $Q^2_{\mathrm{BRST}} = \frac{c-26}{12}c_0$ corresponds to $d^2_{\mathrm{fib}} = \kappa_{\mathrm{tot}}\,\omega_1$ with $\kappa_{\mathrm{tot}} = (c-26)/2$. The functorial capstone assigns to a chirally Koszul boundary algebra $\mathcal{A}$ a holomorphic–topological theory with $\mathrm{Obs}^{\partial} \simeq \mathcal{A}$ and $\mathrm{Obs}^{\mathrm{bulk}} \simeq Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$; its value at $\mathcal{A} = \mathrm{Vir}_c$ under the Brown–Henneaux chart $c = 3\ell/2G_N$ is the boundary reading of pure 3d gravity. This companion states the load-bearing definitions, theorems with their exact hypothesis packages, the physics dictionary, and the genuinely open problems, and places the volume in the five-garden programme.

1 The seed

A three-dimensional field theory on $\mathbb{C}_z \times \mathbb{R}_t$, holomorphic along $\mathbb{C}$ and topological along $\mathbb{R}$, has two inequivalent ways of splitting points. Operators separated along $\mathbb{C}$ collide against the logarithmic kernel $\eta_{ij} = d\log(z_i - z_j)$; the residues of these collisions against operator-product modes assemble into a differential. Operators separated along $\mathbb{R}$ merely order themselves; their fusion assembles into a coproduct. The first structure is the chiral bar differential of Beilinson–Drinfeld chiral algebra theory, realised on the Fulton–MacPherson compactifications $\overline{\mathrm{FM}}_k(\mathbb{C})$; the second is the deconcatenation coproduct on the tensor coalgebra $T^c(s^{-1}\bar{\mathcal{A}})$. A three-dimensional holomorphic–topological theory is precisely a coherent interlocking of the two, and the operad that governs the interlocking –- closed colour $\overline{\mathrm{FM}}_k(\mathbb{C})$, open colour ordered configurations $\mathrm{Conf}^<_m(\mathbb{R})$, mixed colour the Voronov half-plane models $\overline{\mathrm{FM}}_{k,m}(\mathbb{H})$, with the directional restriction $\mathsf{SC}^{\mathrm{ch,top}}(\ldots,\mathsf{op},\ldots;\mathsf{cl}) = \varnothing$ imposed as a zero chain complex –- is the two-coloured Swiss-cheese chiral–topological operad $\mathsf{SC}^{\mathrm{ch,top}}$. Dunn additivity does not apply to it; the decoupled product $\overline{\mathrm{FM}}_k(\mathbb{C}) \times E_1(m)$ is only the depth-zero associated graded; its coupled homotopy-Koszulity is a named conjecture.

Under this identification the standard apparatus of quantization becomes the standard apparatus of homotopical algebra, term by term. The quantum master equation $\hbar\Delta_{\mathrm{BV}}S_{\mathrm{eff}} + \frac{1}{2}\{S_{\mathrm{eff}}, S_{\mathrm{eff}}\}_{\mathrm{BV}} = 0$ is the Maurer–Cartan equation $D_{\mathcal{A}}\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ of the bar construction; at fixed diagrammatic weight it is the planar A_∞ identity, and the A_∞ identity in turn is Stokes' theorem for logarithmic forms on $\overline{\mathrm{FM}}_k(\mathbb{C})$, with the Arnold–Orlik–Solomon relations cancelling the corner strata. The conformal anomaly is bar curvature: the Virasoro ghost system of weights $(2,-1)$ has $c_{\mathrm{gh}} = -26$, and

$$Q^2_{\mathrm{BRST}} = \frac{c-26}{12}\,c_0 \qquad \longleftrightarrow \qquad d^2_{\mathrm{fib}} = \kappa_{\mathrm{tot}} \cdot \omega_1, \qquad \kappa_{\mathrm{tot}} = \kappa_{\mathrm{ch}}(\mathcal{A}) + \kappa_{\mathrm{ch}}(bc) = \frac{c}{2} - 13.$$

Anomaly cancellation $c = 26$ and flatness $\kappa_{\mathrm{tot}} = 0$ are the same equation. Nothing in this dictionary is a metaphor; each entry is either a proved comparison or a comparison problem with a name, and this companion keeps the two apart.

2 The objects

Definition 2.1 (A_∞-chiral algebra)**.** *An A_∞-chiral algebra on a curve is a cochain complex (A, Q) with translation ∂, unit $\mathbf{1}$, and operations $m_k\colon A^{\otimes k} \to A((\lambda_1))\cdots((\lambda_{k-1}))$ of degree $2-k$, the λ_i being spectral parameters Fourier-dual to coordinate differences, packaged as a square-zero degree-one coderivation D_A on $\overline{B}(A) = T^c(s^{-1}\bar{A})$. A curved variant absorbs $m_1^2 \neq 0$ over moduli into m_0. The algebra arises as a chart: $A = A_b = \mathrm{RHom}_C(b,b)$ for a vacuum object b of an open factorization category C; the Morita class of C, not the quasi-isomorphism class of A_b, is the invariant.*

Definition 2.2 (The bicoloured primitive)**.** *The primitive object of the volume is not an algebra but the nine-tuple $(C|_{(X,D,\tau)},\ \mathcal{D}^{\mathrm{cl}}|_X;\ b,\ A_b,\ F^{\mathrm{cl}};\ \beta_{1/2},\ \mathrm{tr}^{\mathrm{cl}}_X;\ \mathrm{Tr}_C)$: open factorization dg-category on a tangential log curve, closed factorization category, boundary vacuum, chart algebra, closed vacuum factorization algebra, half-braiding, closed-colour trace, open-colour cyclic trace. Modularity is a property of the open category –- cyclic trace plus clutching –- whose closed shadow has modular consequences; it is never a property of the closed algebra alone.*

Five objects are never conflated: the chart algebra A_b; the bar coalgebra $\overline{B}(A_b)$, which classifies twisting and coupling; its cohomology A^{i}; the Koszul dual $A^!$, formed by Verdier duality on the Ran space; and the derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \simeq \mathrm{ChirHoch}^\bullet(A_b, A_b)$, which is the bulk. $\Omega\overline{B}(A_b) \simeq A_b$ is bar–cobar *inversion*, not Koszul duality; the bulk is reached by Hochschild cochains, not by the bar.

Definition 2.3 (BV datum)**.** *A 3d holomorphic–topological BV theory on $\mathbb{R}_t \times \mathbb{C}_z$ has field complex $\mathcal{E} = \Omega^\bullet(\mathbb{R}_t)\widehat{\otimes}\Omega^{0,\bullet}(\mathbb{C}_z) \otimes \mathfrak{a}[1]$, differential $Q = d_t + \bar{\partial}_z + d_{\mathfrak{a}}$, odd symplectic pairing $\omega_{\mathrm{BV}}(\alpha,\beta) = \int dz \wedge \langle\alpha,\beta\rangle_{\mathfrak{a}}$, cubic interaction, a Q-compatible Lagrangian boundary condition, and heat-kernel propagator $P_{\epsilon<L} = \int_\epsilon^L Q^* K_s\, ds$ whose singular part is the Cauchy–step kernel $\Theta(t)/2\pi z$.*

Two further objects carry the higher structure. The *modular Swiss-cheese operad* extends the closed colour to all genera, $C_*(\overline{\mathrm{FM}}_k(\Sigma_g))$, with mixed sector the holomorphic–topological screens; its genus-zero restriction is $\mathsf{SC}^{\mathrm{ch,top}}$. The *relative Feynman transform* $\mathrm{FT}_{\mathrm{rel}}(C)$ is the completed stable-graph coalgebra with bicomplex D_0 (genus-preserving bar) and D_1 (modular self-sewing), $(D_0 + D_1)^2 = 0$; it is the exact recipient of all-genus partition data.

3 The theorems

Statuses are the manuscript's own: *proved*, *proved with a fixed associator*, *conditional* on a named package, or *conjecture*. The volume's discipline is that each theorem carries its licensing data in the statement; this companion preserves exactly that.

Theorem 3.1 (Physics bridge; conditional). *A 3d holomorphic–topological QFT on* $\mathbb{R}_t \times \mathbb{C}_z$ *with* (a) *BV datum in product gauge,* (b) *factorized parametrix* $P_{\mathrm{sing}} = \sum_\alpha K_\alpha^{\mathrm{hol}}(z) \boxtimes H_\alpha^{\mathrm{top}}(t)$ *modulo smooth exact terms,* (c) *one-loop finiteness in the sense of Gwilliam–Rabinovich–Williams, and* (d) *product-polynomial interactions, has BV–BRST observables forming a logarithmic* $\mathsf{SC}^{\mathrm{ch,top}}$*-algebra. Hypothesis* (b) *is essential and not automatic: the naive product of Green kernels fails it.*

Theorem 3.2 (Stokes–Arnold calculus; proved). *On the manifold-with-corners* $\overline{\mathrm{FM}}_k(\mathbb{C})$*, closed interior logarithmic forms with operadic boundary factorization obey* $\int_\Gamma d\omega = \sum_{|S|\geq 2} \epsilon_{D_S} \int_{\Gamma\cap D_S} \mathrm{Res}_{D_S}(\omega)$*, with computed orientation signs; the resulting identities on residues are exactly the* A_∞ *identities, the Arnold–Orlik–Solomon relations cancelling all corner contributions.*

Theorem 3.3 (BV equals bar, stratified by class). *The identification of the BV complex with the chiral bar complex,* $Q_{\mathrm{BV}} = \{S, -\} \leftrightarrow d_{\mathrm{bar}}$*, holds: unconditionally at chain level for class* G (Heisenberg, lattice) *and class* L (affine Kac–Moody at non-critical level)*; for class* C ($\beta\gamma$, bc) *conditionally on vanishing of the harmonic-decoupling obstructions; for class* M (Virasoro, $\mathcal{W}_N$) *it is* false *on the raw bounded direct-sum complex and holds in the weight-completed ambient. On cohomology the identification holds for all four classes through the Drinfeld–Sokolov–Hochschild bridge. The interacting obstruction is the harmonic part of the Hodge-decomposed BV propagator* $P_{\mathrm{BV}} = d\log E + [d_{\mathrm{fib}}, h] + P_{\mathrm{harm}}$.

Theorem 3.4 (Genus-zero BRST–bar; conditional). *For a PBW conformal algebra* $\mathcal{A}$ *with* $c = 26$ *there is a quasi-isomorphism* $(\mathcal{A}_{\mathrm{tot}}^{\mathrm{rel}}, Q_{\mathrm{BRST}}) \xrightarrow{\sim} (\overline{B}_0^{\mathrm{ch}}(\mathcal{A}_{\mathrm{tot}}), d_{\mathrm{bar}})$ *by conformal-weight filtration, conditional on vanishing filtered lifting obstructions. Unconditionally,* $\kappa_{\mathrm{tot}} = 0$ *if and only if* $c_{\mathrm{matter}} + c_{\mathrm{ghost}} = 0$*: both squares are controlled by* $c - 26$*. The semi-infinite variant* $\overline{B}^{\mathrm{ch}}(\widehat{\mathfrak{g}}_k) \simeq C^{\infty/2+\bullet}(\widehat{\mathfrak{g}}_k, V_k)$ *holds for all* $k \neq -h^\vee$ *with no anomaly condition –- the semi-infinite wedge absorbs it –- and extends to* $\mathcal{W}$*-algebras by Drinfeld–Sokolov reduction.*

Theorem 3.5 (The $\mathsf{SC}^{\mathrm{ch,top}}$ heptagon; proved with associator). *Seven presentations of* $\mathsf{SC}^{\mathrm{ch,top}}$ *–- coloured operadic; Koszul-dual via distributive laws; factorization on the Ran space; BV/BRST (coderived); convolution* L_∞*; Drinfeld centre of the factorization module category; derived-algebraic-geometric (one-shifted symplectic moduli) –- are pairwise* ∞*-quasi-isomorphic after fixing a Drinfeld associator* $\Phi \in \mathrm{GRT}_1(\mathbb{Q})$*, and the identification is associator-independent on cohomology. Chain-level strictness holds on the evaluation-generated Koszul locus. The arity-*$(2,2)$ *bicoloured coherence polytope is required; the pentagon alone is insufficient. The Swiss-cheese operad is* not *formal* (Livernet)*, so the edge that would follow from formality instead routes through the coupled homotopy-Koszulity of* $\mathsf{SC}^{\mathrm{ch,top}}$ *–- a named conjecture, proved only at the level of associated graded.*

Theorem 3.6 (Chiral higher Deligne). *For* $\mathcal{A}$ *on the Koszul locus:* (1) *the derived chiral centre* $Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}) = \mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ *carries the canonical chiral brace* (E_2-chiral) *action* (proved with associator)*;* (2) *conditional on a signed two-coloured contracting homotopy for* $\Omega((\mathsf{SC}^{\mathrm{ch,top}})^!)$ *–- a genuinely new two-coloured lift of the chiral Positselski theorem, not yet established –- the pair* $(Z_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}), \mathcal{A})$ *is the universal*

$\mathsf{SC}^{\mathrm{ch,top}}$ *bulk–boundary datum;* (3) *with explicit topologization data the mixed structure refines to E_3-topological, associator-dependent at chain level and independent on cohomology. Chiral Hochschild cohomology commutes with Drinfeld–Sokolov reduction:* $\mathrm{ChirHoch}^\bullet(\mathcal{W}_k(\mathfrak{g})) \simeq H^\bullet_{\mathrm{DS}}(\mathrm{ChirHoch}^\bullet(V_k(\mathfrak{g})))$ (weight-completed).

Theorem 3.7 (PVA descent; proved). *The Q-cohomology of a logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra carries a (-1)-shifted Poisson vertex algebra structure: $[a]\cdot[b] = [\mu(a,b)]$ and $\{[a]_\lambda[b]\} = [\sum_n \frac{\lambda^n}{n!} a_{(n)} b]$, with commutativity from the exchange cylinder* (the two line orderings joined through the bulk), *Jacobi from the three-face Stokes argument on $\overline{\mathrm{FM}}_3(\mathbb{C})$, Leibniz from the mixed sector. Vanishing of the higher $m_{k\geq 3}$ on cohomology is* conditional *on compatible topological null-homotopies; contractibility of* $\mathrm{Conf}^<_k(\mathbb{R})$ *alone is not sufficient. Quantization of the PVA is a separate conditional problem* (analytic Knizhnik–Zamolodchikov strong deformation retract, Stokes data, reflected weights, and the lift $T = [Q_{\mathrm{tot}}, G]$).

Theorem 3.8 (Raviolo restriction; proved). *Time-slice restriction of a $\mathsf{SC}^{\mathrm{ch,top}}$-algebra at t_0 glues two formal disks along the punctured disk and produces a dg raviolo vertex algebra, the state–field map recovered from the singular part of m_2 by the divided-power Borel transform $\lambda^n/n! \leftrightarrow z^{-n-1}$; raviolo cohomology carries the Poisson vertex structure.*

Theorem 3.9 (Higher genus; conditional tower). *Dunn additivity fails for $\mathsf{SC}^{\mathrm{ch,top}}$; the genus obstruction is the curvature $d^2_{\mathrm{fib}} = \kappa_{\mathrm{ch}}(\mathcal{A})\,\omega_g$. For chirally Koszul $\mathcal{A}$ at non-critical level and $g \geq 2$, granted modular-bootstrap acyclicity $H^2(\mathfrak{g}^{\mathcal{A}}_{\mathrm{mod}}, d_{\Theta^{(0)}}) = 0$ and the degree-two comparison criterion, the twisting cochain extends genus by genus and the modular bar complex acquires a twisted-tensor representative; Fay's trisecant identity supplies the genus-g Arnold relations for the prime-form propagator. The complete modular bar object* is *the relative Feynman transform* (proved, and explicitly formal: the stable-graph skeleton does not determine global periods or monodromy).

Theorem 3.10 (Unified chiral quantum group; proved on its verified surface). *For simple* $\mathfrak{g}$, *a good grading from an $\mathfrak{sl}_2$-triple, level $k \neq -h^\vee$ and shift μ, there is a chiral bialgebra datum $Q^{k,f,\mu}_{\mathfrak{g}}$ with shifted-parameter coassociative spectral coproduct, spectral R-matrix satisfying the Yang–Baxter equation with quasi-triangularity $R(z)\Delta^{\mathrm{op}}_z = \Delta_z R(z)$, Koszul dual realising bar–cobar inversion, and Drinfeld–Sokolov specialisation $Q^{k,f_{\mathrm{prin}},0}_{\mathfrak{g}} = \mathcal{W}_k(\mathfrak{g})$. The verified surface is the locus where five named obstruction classes vanish. Specialisation fibres include the Yangian, affine and shifted Yangians, truncated BFN Coulomb-branch algebras, finite and affine $\mathcal{W}$-algebras. The super-chiral Yangian satisfies $(Y^s_\hbar(\mathfrak{g}))^! \simeq Y^s_{-\hbar}(\mathfrak{g})^\theta$ with the Chevalley–Kac involution, at all orders via the unitarity functional equation $R_s^{-1}(u) = R_s(-u-\hbar h_s^\vee)$.*

Theorem 3.11 (Ordered associative chiral Koszul duality; proved at genus zero). *For a strongly admissible augmented associative chiral algebra A with bar coalgebra $C = \overline{B}^{\mathrm{ch}}(A)$: the ordered chiral shuffle gives $\overline{B}(A \otimes^{\mathrm{ch}} B) \simeq \overline{B}(A)\widehat{\otimes}\overline{B}(B)$; opposite algebras give co-opposite coalgebras; there is a PBW-complete equivalence of bimodule and bicomodule derived categories; the diagonal bimodule corresponds to the coalgebra C_Δ –- the diagonal is the universal boundary condition –- and is the unit for derived cotensor; Hochschild and coHochschild theories are dual; and when A is d-Calabi–Yau the ordered pair-of-pants and trace operations carry a d-shifted ordered Frobenius structure.*

Theorem 3.12 (Fractional ghosts; proved). *Non-principal Drinfeld–Sokolov reductions carry half-integral Kazhdan gradings; a Kummer branched cover of the curve of degree d_f integralises the grading, the Sugawara antighost identity $[Q, G'] = T$ holds on the cover and Galois-descends. Consequently the* *derived chiral centres of the Bershadsky–Polyakov algebra* $\mathcal{W}_3^{(2)}$*, of minimal* $\mathcal{W}^k(\mathfrak{sl}_4, f_{\min})$*, and of* all *good-graded nilpotent reductions carry* E_3*-topological structure –- the topologisation ladder extends unconditionally beyond principal gradings.*

Theorem 3.13 (Universal Holography; coherence theorem on realization data)**.** *There is a functor* $\Phi_{\mathrm{hol}} : \mathsf{ChirAlg}_X^{\omega,\mathrm{BL},\mathrm{adm},\Xi} \longrightarrow \mathsf{HT\text{-}QFT}_{X\times\mathbb{R}}^{\Xi}$ *on chirally Koszul algebras with conformal vector at non-critical level, boundary linearisation, and holomorphic–topological realization datum* Ξ*, such that* $\mathrm{Obs}^{\partial}(T_{\mathcal{A}}) \simeq \mathcal{A}$ *as* E_1*-chiral algebras and* $\mathrm{Obs}^{\mathrm{bulk}}(T_{\mathcal{A}}) \simeq Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ *as* E_3*-topological factorization algebras –- on the ordinary complex for classes* $\mathsf{G}/\mathsf{L}/\mathsf{C}$*, weight-completed for class* M*. The realization datum* Ξ contains *the physical Costello–Gwilliam model and both comparison maps; without* Ξ *the construction supplies only the algebraic centre candidate. The five rungs are: Heisenberg* (abelian Chern–Simons)*, affine Kac–Moody,* Vir_c (the 3d-gravity rung at Brown–Henneaux $c = 3\ell/2G_N$)*,* $\mathcal{W}_{N,c}$ (Casimir tower, E_{N+1})*, and* $\mathcal{W}_\infty[\lambda]$ (the E_∞ endpoint, conditional on the Prochazka, Creutzig–Kanade–Linshaw, Pope–Romans–Shen/Bakas and Yamada hypotheses).

Theorem 3.14 (Ten projections of the Maurer–Cartan equation)**.** *For a chirally Koszul, cyclically admissible* $\mathcal{A}$*, the element* $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0 \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ *projects to:* (1) *genus-by-genus* finiteness *of* $\langle \Theta_{\mathcal{A}}^{\otimes n}\rangle_g$ (integrals of cohomology classes over compact $\overline{\mathcal{M}}_{g,n}$; on the scalar lane the genus series has radius $4\pi^2$ and closed form $\kappa\big(\frac{\sqrt{\hbar}/2}{\sin(\sqrt{\hbar}/2)} - 1\big)$)*;* (2) complexity $r_{\max}$ *counting nontrivial bulk brackets;* (3) *a* $-(3g-3)$*-shifted symplectic structure with complementary Lagrangians* $\mathbf{Q}_g(\mathcal{A})$*,* $\mathbf{Q}_g(\mathcal{A}^!)$ *and the Verdier involution as anti-symplectomorphism;* (4) S-duality $\sigma(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}^!}$ *with conductor sums* $c + c' = 2\dim\mathfrak{g}$ *for* $\widehat{\mathfrak{g}}_k$ *along* $k \mapsto -k-2h^\vee$ *and* $K_N = 4N^3 - 2N - 2$ *for* $\mathcal{W}_N$ $(26, 100, 246)$*;* (5) *the classical Yang–Baxter equation for the collision residue* $r(z)$ (the quantum-group upgrade beyond the affine lineage is a conjecture)*;* (6) *soft residues: the degree-two Ward identity* $S^{(0)}_{\mathrm{cel}} = \sum_i \kappa(\phi_s, \phi_i)/(z_s - z_i)$ (the Weinberg reading requires the celestial transform, asymptotic-state comparison, and BMS normalisation –- a separate comparison problem)*;* (7) *the* modular bootstrap *recursion* $d_{\Theta^{(0)}}\Theta^{(g)} = -\frac{1}{2}\sum_{g_1+g_2=g}[\Theta^{(g_1)}, \Theta^{(g_2)}]$*, determining each genus from lower genera up to gauge;* (8) *algorithmic* reconstruction *of the six-component holographic datum from boundary OPE data, with holographic code distance equal to Verdier distance* (∞ for Heisenberg; 3 for the Lee–Yang algebra $\mathrm{Vir}_{-22/5}$, matching the $[[5,1,3]]$ pentagon code)*;* (9) *the* critical dichotomy*: self-duality of the Virasoro family at* $c = 13$*, maximal asymmetry at* $c = 26$ *where the dual* Vir_0 *is uncurved and every* F_g *vanishes by matter–ghost cancellation;* (10) Fredholm form *of the partition function on the Gaussian lane:* $Z_1(\mathcal{H}_\kappa;\tau) = (\mathrm{Im}\,\tau)^{-\kappa/2}|\eta(\tau)|^{-2\kappa}$ *and* $Z_g = \det(1-K_g)^{-\kappa}$ *with* K_g *the Bergman sewing kernel; for class* M *the nonvanishing higher cumulants obstruct any single Fredholm determinant* (proved obstruction).

Theorem 3.15 (3d gravity, honestly)**.** *At* $\mathcal{A} = \mathrm{Vir}_c$*: the same-family Koszul representative is* Vir_{26-c}*, with* $\kappa + \kappa' = 13$ *fixed at* $c = 13$ (on the strict comparison surface)*. The effective genus tower after ghost subtraction is* $F_g^{\mathrm{eff}} = \kappa_{\mathrm{eff}}\lambda_g^{\mathrm{FP}}$*,* $\sum_{g\ge 1} F_g x^{2g} = \frac{c-26}{2}(\hat{A}(ix) - 1)$*, all* $F_g = 0$ *at* $c = 26$*. The Brown–Henneaux identification* $c = 3\ell/2G_N$ *is an external chart* (proved elsewhere)*. The identification of line operators with highest-weight* Vir_{26-c}*-modules –- conical defects of deficit* $2\pi(1 - \sqrt{1 - 8G_N h})$ *–- is a* conjecture*. The Cardy formula* $S = 2\pi\sqrt{c\Delta_+/6} + 2\pi\sqrt{c\Delta_-/6}$ *and the Bekenstein–Hawking form* $S_{\mathrm{BTZ}} = 2\pi r_+/4G_N$ *are* conjectures *for the bar-complex trace: the proved input is the vacuum energy* $\kappa = c/2$*; modular invariance, vacuum dominance and saddle extraction are imported hypotheses. The Maloney–Witten-type completion –- the* $SL_2(\mathbb{Z})$ *sum equalling the Borel–Zwegers completion of the raw bar trace –- is conditional on resurgence hypotheses; its unconditional shadow at* $c = 24$ *is the automorphic* *identity extracting the partition function from* $-\log \Phi_{10}^{\mathrm{un}}$*, the Igusa form entering Volume II from the gravity side.*

Theorem 3.16 (Celestial identifications and anti-identifications)**.** *For a perturbative BV theory on* $\mathbb{R}^4$ *with twist data, transverse boundary condition, vanishing boundary anomaly and gluing class, the celestial boundary chiral algebra exists functorially, the pair* $(\mathcal{A}^{\mathrm{cel}}, Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}^{\mathrm{cel}}))$ *is canonically an* $\mathsf{SC}^{\mathrm{ch,top}}$*-algebra, and the associated graded of the celestial OPE is chiral factorization homology; exactness of the filtered comparison is equivalent to vanishing Mellin-cone obstructions. Soft theorems are bar cohomology classes: the* g*-loop sub*$^{r-2}$*-leading soft theorem lives in* $H^{r-1}(\overline{B}^{\mathrm{ch}}(\mathcal{A}^{\mathrm{cel}}))$*. Two proved* anti-*identifications police the dictionary: the affine bar residue is the level channel* $k\kappa^{ab}/z$*, not the physical celestial OPE coefficient* $B(\Delta_1{-}1, \Delta_2{-}1)f^{ab}{}_c/z_{12}$*; the Virasoro collision kernel* $(c/2)/z^3 + 2T/z$ *has a central triple pole that the physical graviton OPE does not. Helicity is chirality: the bar complex is the self-dual sector, the Koszul dual the anti-self-dual, and the maximally-helicity-violating amplitude is the minimal Verdier pairing.*

4 The dictionary and the constants

The physics dictionary, each entry either proved or named as a comparison problem:

Physics	**Mathematics**
fields $\phi \in \mathcal{A}$	bar insertions at marked points of $\overline{C}_{n+1}(X)$
antifields ϕ^+	Verdier–Serre dual, shifted by one
field–antifield flip	Verdier duality $\mathbb{D}_{\mathrm{Ran}}$
ghost bilinear $c_0c_1\lvert 0\rangle$	the propagator $\eta_{12} = d\log(z_1 - z_2)$
ghost complex	$\overline{B}_X(\mathcal{A}^!)$
BV antibracket	Poincaré residue along collision divisors
BRST differential	bar differential (transferred equality through the boundary retract)
quantum master equation	Maurer–Cartan = A_∞ = Stokes
BV Laplacian	diagonal-residue loop insertion (heuristic at chain level)
anomaly $Q^2 = \frac{c-26}{12}c_0$	curvature $d^2_{\mathrm{fib}} = \kappa_{\mathrm{tot}}\omega_1$, $\kappa_{\mathrm{tot}} = \frac{c-26}{2}$
gauge coupling	level k; $\quad \hbar = 1/(k+h^\vee)$
line operators	$\mathcal{A}^!$-modules
S-duality	Koszul duality, on the coupling-independent locus
BPS spectrum	$H^*(\overline{B}(\mathcal{A}))$
Coulomb branch	$\mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$
genus-g partition function	$\Theta^{(g)}_{\mathcal{A}}$; scalar lane $F_g = \kappa\lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$

The recurring constants. $\kappa_{\mathrm{ch}}(\mathrm{Vir}_c) = c/2$; $\kappa_{\mathrm{ch}}(bc_{\lambda=2}) = -13$; $F_1 = \kappa/24$; the Faber–Pandharipande values $\lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}}\frac{|B_{2g}|}{(2g)!}$ give $F_1 = \frac{\kappa}{24}$, $F_2 = \frac{7\kappa}{5760}$, $F_3 = \frac{31\kappa}{967680}$. The derived-centre complementarity $\rho_K = \kappa_{\mathrm{ch}}(\mathcal{A}) + \kappa_{\mathrm{ch}}(\mathcal{A}^!)$ takes the computed values 0 on classes $\mathsf{G}/\mathsf{L}/\mathsf{C}$, 13 on Virasoro, 250/3 on $\mathcal{W}_3$. For Bershadsky–Polyakov the proved identity is the central-charge conductor: with $c_{\mathrm{BP}}(k) = -(2k+3)(3k+1)/(k+3)$ one has $c_{\mathrm{BP}}(k) + c_{\mathrm{BP}}(-k-6) \equiv 50$, self-conjugate value $c^* = 25$ at $k = -3 \pm 2i$, image

$c \le 1$ for $k > -3$ and $c \ge 49$ for $k < -3$; on the anomaly-ratio lane $\varrho_{\mathrm{BP}} = 1/6$ this gives the conditional value $\rho_K(\mathrm{BP}) = 25/3$, pending the genus-one curvature computation of κ_{BP} itself (the once-recorded values $K = 196$ and $98/3$ are retracted). The Mukai conductor $K^\kappa = 2c_+(\mathrm{II}_{4,20}) = 8 = \mathrm{ord}(H_1)$ on the $K3$ row, with $\hbar^2 K^\kappa = -1$, is a separate sixth invariant, not a ρ_K value. The S-duality discrepancy $\Delta(\Psi, N) = \frac{N^2+2N-1}{2N}(\Psi + \Psi^{-1})$ measures exactly where "S-duality equals Koszul duality" fails at finite coupling; the Langlands form is $\mathcal{A}^!(\widehat{\mathfrak{g}}_k) \simeq \mathcal{A}(\widehat{{}^L\mathfrak{g}}_{k^\vee})$, $k^\vee = -h^\vee_{{}^LG} + 1/k$.

The seven faces of the collision residue $r(z)$ –- bar–cobar twisting morphism; line-operator Maurer–Cartan element; Poisson vertex λ-bracket; commuting sphere Hamiltonians; Drinfeld Yangian r-matrix; Sklyanin bracket; Gaudin Hamiltonians –- are identified under the hypotheses of the seven-face master theorem, with $\hbar = 1/(k + h^\vee)$ throughout; the set of presentations is a torsor over $\mathrm{GRT}_1(\mathbb{Q})$.

5 The garden

One logarithmic 1-form generates the programme. On a smooth complex curve X, the form $\eta = d\log(z_1 - z_2)$ is the unique coordinate-free 1-form with a residue along the diagonal of $\mathrm{Conf}_2(X)$; the Arnold relation $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ generates all relations among its higher companions, and its failure of coordinate-invariance is the cocycle $d\log w'(z)$ that the Virasoro algebra measures. From η one builds, for an augmented chiral algebra $\mathcal{A}$ on X, the ordered bar complex $\overline{B}^{\mathrm{ord}}_X(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) \otimes \Omega^\bullet_{\log}(\overline{\mathrm{FM}}_\bullet(X))$; its differential squares to zero exactly when $\mathcal{A}$ satisfies the Borcherds identity, so the bar complex is synonymous with the chiral-algebra axiom. Assembled over $\overline{\mathcal{M}}_{g,n}$ with the prime-form propagator, the total differential is a Maurer–Cartan element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$; genus ≥ 1 flatness fails by the curvature $d^2_{\mathrm{fib}} = \kappa(\mathcal{A})\,\omega_g$, and the scalar κ –- $\kappa(\mathrm{Vir}_c) = c/2$, $\kappa(V_k(\mathfrak{g})) = \dim\mathfrak{g}\,(k + h^\vee)/2h^\vee$, $\kappa(\mathcal{H}_k) = k$, $\kappa(\mathcal{W}_N) = c(H_N - 1)$ –- is the central charge read as modular curvature. The Verdier–Koszul dual obeys conductor laws $c(\mathcal{A}) + c(\mathcal{A}^!) = K$ with $K(\mathrm{Vir}) = 26$, $K(\mathcal{W}_N) = 4N^3 - 2N - 2$, $K(V_k(\mathfrak{g})) = 2\dim\mathfrak{g}$ along $k \mapsto -k - 2h^\vee$; self-dual and uncurved points organise the landscape into the archetype classes `G/L/C/M` of shadow depth $2/3/4/\infty$ plus the critical Feigin–Frenkel stratum.

Every object sits on one chain: primitive factorization category $\to$ chart algebra $A_b \to$ bar shadow $\overline{B}(A_b) \to$ derived chiral centre $Z^{\mathrm{der}}_{\mathrm{ch}}(A_b)$ (the bulk) $\to$ acting line operators $\to$ scalar modular shadow. Identifying stages without the licensing datum –- shadow = object –- is the master error the programme is disciplined against.

The five gardens are five readings of the chain. *Volume I* (`chiral-bar-cobar`) builds the engine on algebraic curves: the bar–cobar adjunction and inversion, quadratic Koszul recognition, centre complementarity with its shifted symplectic structure, the deformation–Hodge–graph traces $F_g = \kappa\lambda_g^{\mathrm{FP}} + \delta F_g^{\mathrm{cross}}$, Hochschild concentration, and the arithmetic shadow of the tower. *This volume* reads the same object as physics: bar = BV/BRST, quantum master equation = Maurer–Cartan, anomaly = curvature, $\mathsf{SC}^{\mathrm{ch,top}}$ as the universal bulk–boundary home, Universal Holography $\mathcal{A} \mapsto (\partial = \mathcal{A},\ \mathrm{bulk} = Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}))$, and at $\mathrm{Vir}_{c=3\ell/2G_N}$ the boundary reading of pure 3d gravity. *Volume III* (`calabi-yau-quantum-groups`) constructs the geometric source: the two-stage functor $\Phi_d = \mathrm{Sp}^{\mathrm{ch}}_{\Sigma_{d-1},C} \circ \Phi^{\mathrm{FA}}_d$ from Calabi–Yau d-categories through holomorphic factorization algebras to chiral algebras on a curve, the cohomological Hall algebra $Y^+(X)$ with Drinfeld double $G(X)$ as the level-three quantum group, and the κ-tuple $(\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}^{\mathrm{Hodge}}, \kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}) = (0, 0, 3, 5, 24)$ at $K3 \times E$. *Mixed holomorphic–topological strings* computes the local germ of the $d = 2$ theory: at N Dirac branes on $\mathbb{R}^2_{\mathrm{top}} \times \mathbb{C}^2_{\mathrm{hol}}$ the trace $J(f) = \mathrm{Tr}\,f(\phi_1, \phi_2)$ is the formal-Darboux coordinate of the trace sector, with a typed ledger separating the local stalk from the global theory. *The Igusa cusp form* (`igusa-cusp-form`) is the terminal scalar: $\Delta_5 = \Phi_{10}^{1/2}$, weight 5 on $\mathrm{Sp}_4(\mathbb{Z})$, read five ways –- automorphic section, Borcherds–Kac–Moody denominator, protected Pfaffian, mirror discriminant, singular theta lift of $\phi_{0,1}$ –- with $Z_{\mathrm{BPS}}^{K3\times E} = \Delta_5^{-2}$ the level-five shadow of the $K3 \times E$ object.

The single question underneath: characterise the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ as a geometric object. The proposed answer –- endomorphisms of the Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\mathrm{vac}}(\mathcal{A})$ in a (-2)-shifted symplectic stack of vacua –- has Volume I computing its Taylor jets, this volume its local composition law, Volume III its geometric source, and the satellites its sharpest example: the $K3 \times E$ datum whose scalar shadow is Δ_5.

6 The frontier

Four construction problems carry the open weight of the volume.

(i) *The $K3 \times E$ operator package.* Construct $\mathfrak{D}_X \in \mathrm{End}_{\mathrm{ChirHoch}}(H_X)$ on $K3 \times E$ with protected Pfaffian Δ_5 –- the operator-level object whose scalar shadow the Igusa volume studies.

(ii) *The gravity line.* Construct the line-operator algebra acting on the boundary whose scalar trace is $\Phi_{10}^{\mathrm{un}} = \Delta_5^2$ –- the level-A completion of the $K3 \times E$ story.

(iii) *PVA quantization.* Promote the classical (-1)-shifted Poisson vertex structure to the all-loop quantum theory: analytic KZ strong deformation retract, Stokes data, reflected weights, and the lift $T = [Q_{\mathrm{tot}}, G]$.

(iv) *Two-coloured chiral Positselski.* Establish the signed two-coloured contracting homotopy that upgrades the chiral higher Deligne theorem from brace action to universal bulk–boundary datum.

Behind these stand the structural conjectures: coupled homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ (Livernet nonformality blocks the easy route); the factorized parametrix hypothesis of the physics bridge; class-M chain-level structure beyond the weight-completed ambient (provably false on the raw direct-sum complex); the $\mathcal{W}_\infty[\lambda] \Rightarrow E_\infty$ endpoint beyond its four-hypothesis window; and the analytic hypotheses –- modular invariance, vacuum dominance, saddle and resurgence data –- that separate the proved algebraic genus tower from Cardy, BTZ and Maloney–Witten physics. The honest summary: the volume proves a rich algebraic theory of the bar complex whose scalar sector reproduces the known one-loop normalisations of 3d gravity and whose structure maps mirror the holographic dictionary; that this theory *is* quantum gravity is precisely the content of the named open comparisons, no more and no less.